[INCH-POUND]
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COMMERCIAL ITEM DESCRIPTION

DRILLING MACHINE, RADIAL, BENCH-MOUNTED, SLIDING ARM AND SWIVEL-HEAD, HAND-FEED

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers hand-feed, sliding arm and swivel-head, bench-mounted, radial drilling machines intended for use in drilling, tapping, and reaming operations in machine shops, ground installations, and mobile repair units where special horizontal or vertical radial drilling operations are encountered.

2. CLASSIFICATION. The radial drilling machines shall be of the following types. The type to be furnished shall be as specified (see 7.2(b)).

Type I - Adjustable head, fixed worktable
Type II - Fixed head, adjustable worktable

3. SALIENT CHARACTERISTICS

3.1 General requirements. The machine shall be new and one of the manufacturer's current models capable of operation in accordance with the requirements herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2 Components. The radial drilling machines shall include, as a minimum, the following components:

3.2.1 Base. The base shall be of sufficient strength and rigidity to withstand all vibration, stress, and loads from normal operations. Means shall be provided for securing the base to a bench, table, or stand. When specified (see 7.2(c)), the manufacturer shall supply a table or stand. The type II machine base shall serve as an auxiliary worktable and shall have a means to secure the workpieces.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data which may improve this document should be sent to: Defense Supply Center Richmond (DSCR), ATTN: DSCR-VBD, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610.

AMSC N/A FSC 3413

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3.2.2 Worktable. The worktable shall have a means to secure workpiece(s) and shall have sufficient strength and rigidity to support the workpiece(s) during drilling operations. The number and size of T-slots shall be as specified (see 7.2(d)). The dimensions of the T-slots shall conform to the requirements of American National Standards Institute (ANSI) B5.1, "T-Slots, Their Bolts, Nuts and Tongues". Type I machine worktables shall have a coolant trough and drain. Type II machine worktables shall be vertically adjustable and shall rotate to allow access to the auxiliary worktable.

3.2.3 Column. The column shall support the cradle and its associated components. The ability to lock the cradle to the column shall be provided. Type I machines shall have an adjustable column to set the drill head and ram height.

3.2.4 Cradle. The cradle shall support the ram, drill head, and motor. The cradle shall be attached to the column and allow the ram to move radially and swivel 360 degrees horizontally. The cradle shall lock the ram in position to prevent movement after adjustment.

3.2.5 Ram. The ram shall carry and provide support to the drill head and motor. The swing for type I and II rams shall be a minimum of 54 and 14 inches, respectively. Ram adjustment shall be by hand.

3.2.6 Drill head. The drill head assembly shall support the spindle and quill. The drill head shall lock in place after tilt adjustment. Type I machines shall have a minimum tilt of 45-degrees to the left and 30 degrees to the right. Type II machines shall have a minimum tilt of 90 degrees to the left and 45 degrees to the right.

3.2.7 Spindle and quill assembly. The spindle and quill assembly on type I and type II machines shall be moved by hand feed with a minimum travel of 6 inches and 3 inches, respectively. The spindle and quill assembly shall maintain alignment throughout the travel and shall be counterbalanced and automatically retract when the feed is released. Unless otherwise specified (see 7.2(e)), the spindle speeds and steps shall be those considered standard by the manufacturer.

3.2.7.1 Chuck. Unless otherwise specified (see 7.2(f)), the chuck shall have a drill bit diameter range of 0 to 1/2 inch. The chuck shall either be a key chuck or a keyless chuck as specified (see 7.2(g)).

3.2.7.2 Depth stop. Type I machines shall have a depth stop with a positive locking device. The depth stop shall have an indicating scale with a minimum increment of not more than 1/16 of an inch.

3.2.8 Controls. The machine shall be equipped with an on/off power switch. The switch shall be located for convenient access to an operator standing in front of the machine.

3.2.9 Lubrication system. Machine parts requiring periodic manual lubrication shall be readily accessible.

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3.2.10 Coolant system. Type I machines shall have a coolant system. Unless otherwise specified (see 7.2(h)), cutting tools shall be flood cooled, and the machine shall be equipped with a system to control the rate of coolant flow from the reservoir and to direct coolant flow to nozzles mounted on the spindle head. The system shall have a means for cleaning or straining the coolant.

3.2.11 Electrical system. The electrical system shall conform to the ANSI/National Fire Protection Association (NFPA) 79, "Machinery, Industrial, Electrical Standards For". The machine shall operate from a 115-volt, single-phase, 60-hertz circuit. Unless otherwise specified (see 7.2(i)), type I machines shall operate from a dual 115/230-volt, single-phase, 60-hertz circuit and shall be wired for 115-volt operation. Unless otherwise specified (see 7.2(j)), the control system shall operate on a 115-volt, single-phase, 60-hertz circuit.

3.2.12. Electric motors. The drill motor(s) shall have sealed and permanently lubricated ball or roller bearings, shall be energy efficient, and shall conform to the National Electrical Manufacturers Association (NEMA) MG-1, "Motors and Generators". The motor controls shall conform to NEMA ICS-l, “Industrial Control and Systems, General Standards For”. Unless otherwise specified (see 7.2(k)), the motors shall have open dripproof enclosures.

3.2.13 Optional accessories. Optional accessories shall be provided as specified (see 7.2(1)).

3.3 Performance. Unless otherwise specified (see 7.2(m)), the radial drilling machine shall meet the alignment requirements shown in table I and the performance requirements shown in table II when tested in accordance with sections 5.3 through 5.3.4.

TABLE I. Alignment requirements.

Parameter	Tolerance
Spindle runout	
Spindle retracted	0.003 inch
Spindle fully extended (or at 6 inches)	0.005 inch
Spindle travel square to base	0.012 inch/foot
Arm vertical travel square to base (type I)	0.017 inch/foot
Table travel square to column	0.017 inch/foot

TABLE II. Performance requirements.

Parameter	Tolerance
Vertical alignment	0.010 inch/foot
Hole straightness	0.001 inch from hole top to bottom
Hole maximum diameter	Test drill diameter + 0.007 inch
Out of roundness	0.003 inch

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3.4 Machine dimensions and weight. If required, the maximum machine dimensions (length, width, and height) and machine weight shall not exceed the restrictions specified (see 7.2(n)).

3.5 Safety and health requirements. The machine shall be designed and manufactured in accordance with ANSI B11.8, "Machines, Drilling, Milling and Boring, Safety Requirements for the Construction, Care and Use Of". The manufacturer shall ensure that the machine and all equipment and accessories used on the machine shall be in compliance with Occupational Safety and Health Administration (OSHA) 29 CFR PART 1910, "Occupational Safety and Health Standards for General Industry". If a conflict arises between the ANSI B11.8 and OSHA standards, the OSHA standards shall apply.

3.6 System of units. The U.S. Customary System of Units (US) or the International System of Units (Sl) shall be used to graduate measuring and indicating devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 7.2(o)). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system. When specified (see 7.2(p)), measuring and indicating devices shall be graduated in both the US and SI system of measurements.

3.7 Nameplate. A nameplate shall be securely attached to the machine. Unless otherwise specified (see 7.2(q)), the nameplate shall contain the following information:

a. Nomenclature
b. Manufacturer’s name
c. Serial number
d. Machine model designation
e. Power input (volts, total amperes, phase, frequency)
f. Short-circuit/over-current rating
g. Contract number or order number
h. National stock number
i. Date of manufacture

3.8 Lubrication plate or chart. When specified (see 7.2(r)), a lubrication plate or chart shall be attached to the machine. The information provided on the plate or chart shall include:

a. Points of lubricant application
b. Servicing interval
c. Type of lubricant(s) with SAE number or lubricant identifier

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Environmental protection. The item shall meet all applicable Environmental Protection Agency (EPA) restrictions in effect on the date of the contract. These regulations apply to the

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emission of materials hazardous to the environment or the user's health and shall be adhered to during the manufacturing, service, transportation, storage, and operation/use of the item.

5. QUALITY ASSURANCE PROVISIONS

5.1 Product conformance. The products shall meet the salient characteristics of this commercial item description; conform to the manufacturer's own drawings, specifications, standards, and quality assurance practices; and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

5.2 Inspection. The radial drilling machines shall be inspected to determine compliance with all requirements specified in this CID.

5.3 Testing. Unless otherwise specified (see 7.2(s)), the radial drilling machine shall be tested in accordance with sections 5.3.1 through 5.3.4.

5.3.1 Test setup and conditions. Unless otherwise specified (see 7.2(t)), the manufacturer shall be responsible for supplying all tooling and material(s) and shall conduct all tests required by the Government. The manufacturer shall provide test instruments and equipment with calibration traceable to National Institute of Standards and Technology (NIST) and shall provide certification of calibration as specified (see 7.2(u)). Unless otherwise specified (see 7.2(v)), the machine shall be tested in accordance with the following environmental conditions: The ambient temperature shall be between 41 degrees and 104 degrees Fahrenheit and the humidity shall be between 20 percent and 95 percent.

5.3.2 Operational test. The radial drilling machine shall be operated in accordance with the manufacturer's standard operating procedures. The radial drilling machine shall be operated at no load for not less than l/2 hour. Proper operation of all controls and features shall be verified during the trial period.

5.3.3 Performance tests. The performance tests shall be conducted in accordance with sections 5.3.3.1 through 5.3.3.2. There shall be no evidence of overloading, overheating, excessive vibration, or deformation of any part of the machine during or after the test.

5.3.3.1 Alignment test. A machine alignment test shall be performed before and after the drill test using the criteria shown in table I.

5.3.3.2 Drill test. A test workpiece 1.0 inch thick and made of cast iron with a Brinell hardness of 120-150 shall be used. The test workpiece shall be large enough to perform all the required tests. The drill head shall be positioned at its maximum swing position throughout the test. Five holes shall be drilled through the test workpiece. All drilling shall be finished in 12 minutes. The drill bit nominal diameter shall be 1/2 inch. Drill bit material, drill spindle speed, and feed rate shall be at the option of the manufacturer. The holes shall be measured and results shall meet the performance requirements shown in table II. The diameter of all holes shall be measured in three places; the top, middle, and bottom.

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5.3.4 Optional tests. Optional tests shall be conducted as specified (see 7.2(w)).

5.4 Acceptance. Unless otherwise specified (see 7.2(x)), preliminary and final acceptance tests shall be conducted at the manufacturer's site. The results of the alignment and performance tests shall be compared with the requirements shown in tables I and II, respectively. Failure of the machine to meet these requirements shall be cause for rejection.

6. PACKAGING.

6.1 Preservation, packing, and marking. For acquisition purposes, the products shall be preserved, packed, and marked as specified in the acquisition order (see 7.2(y)).

7. NOTES

7.1 Sources of documents

7.1.1 FAR. The FAR may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.

7.1.2 ANSI standards. Copies of ANSI standards may be obtained from the American National Standards Institute, 11 West 42nd Street, New York, NY 10036.

7.1.3 NFPA standards. Copies of NFPA standards may be obtained from the National Fire Protection Association, 1 Batterymarch Park, Quincy, MA 02269-9101.

7.1.4 NEMA standards. Copies of NEMA standards may be obtained from the National Electrical Manufacturers Association, 1300 North 17th Street, Suite 1847, Rosslyn, VA 22209.

7.1.5 OSHA standards. Copies of OSHA standards may be obtained from the U.S. Department of Labor, 200 Constitution Avenue NW, Room 423, Washington, DC 20210.

7.2 Ordering data. Acquisition documents must specify the following information:

a. Title, number, and date of this document
b. Type required (see 2)
c. Base, if required (see 3.2.1)
d. Number and size of worktable T-slots, (see 3.2.2)
e. Spindle speeds, if different (see 3.2.7)
f. Chuck drill bit diameter range, as specified (see 3.2.7.1)
g. Style of chuck, as specified (see 3.2.7.1)
h. Coolant system, if different (see 3.2.10)
i. Electrical system, if different (see 3.2.11)
j. Control voltage, if different (see 3.2.11)
k. Electric motor, if different (see 3.2.12)
l. Optional accessories, as specified (see 3.2.13)
m. Performance, if different (see 3.3)
n. Machine dimensions and weight, if required (see 3.4)

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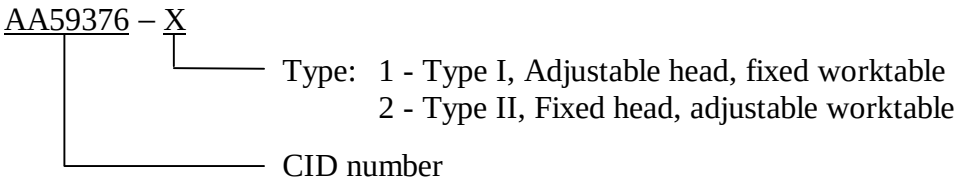
o. System of units required (see 3.6)
p. Dual system of units, if required (see 3.6)
q. Nameplate, if different (see 3.7)
r. Lubrication plate or chart, if required (see 3.8)
s. Testing, if different (see 5.3)
t. Test responsibility, if different (see 5.3.1)
u. Test equipment calibration, if required (see 5.3.1)
v. Test environmental conditions, if different (see 5.3.1)
w. Optional tests, as specified (see 5.3.4)
x. Acceptance, if different (see 5.4)
y. Packaging requirements, as specified (see 6.1)

7.2 Part identification number (PIN). The following part identification numbering system is for government purposes and does not constitute a requirement for the contractor.



MILITARY INTERESTS:

Custodians:
Air Force - 99
Army - AL
Navy - SH

Reviewers:
Air Force - 84
Army - AV, SM
Navy - OS

CIVIL AGENCY
COORDINATING ACTIVITY:

GSA - 6FET

Preparing activity:
DLA - GS

(Project 3413-0165)

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